

Current MoE Reasoning Experiments

1 Scope

This report tracks the three active workflows: `gepaLLMAsJudge`, `probeTest`, and `correlation_pipeline`. The first two were called Experiments 0a and 0b in the previous combined manuscript. Their full protocols, intermediate results, tables, limitations, and external-validation results are retained here under their current names.

The previous combined manuscript for numbered Experiments 1–5 is preserved at `old/report/legacy_full_report.tex` together with its compiled PDF. Those archived results remain available for provenance but are not mixed into the current experimental plan.

2 Experimental Setup

Current experimental scope. The active repository contains three complementary Qwen workflows. `gepaLLMAsJudge` is an annotation-capability study that optimizes Qwen3.6-27B and Qwen3.8-27B prompts for sentence-level Schoenfeld episode classification. It does not provide evidence about Qwen router dynamics. `probeTest` is a representation-probe replication that teacher-forces the 38 released DeepSeek-R1 Schoenfeld traces through the official GPTQ 4-bit Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 checkpoint and tests whether the next sentence’s label is linearly decodable at its causal pre-sentence boundary. The separate `correlation_pipeline` generates benchmark solutions with Qwen3.5-35B-A3B, replays them through the matching checkpoint, and relates router and hidden-state summaries to correctness and reasoning-event labels.

GEPA judge design. The GEPA workflow starts from 3,125 annotated units in 38 released DeepSeek-R1 SAT responses. Complete responses are assigned to partitions before their units are flattened, preventing neighboring units from crossing splits. A local `llama.cpp` server supplies Qwen predictions. Each request contains the SAT problem and the previous, current, and next response units. The original context-aware seed instruction contains 21 audited synthetic contrastive demonstrations. The Qwen3.8 alignment instead imports the 40-example prompt selected by the earlier final fit as its external seed, so it does not dynamically sample new demonstrations. The implementation supports inverse-frequency weighted exact match and an informative same-model LLM-as-a-judge reward. Prompt-selection rules are evaluated with nested response-grouped cross-validation; the final-fit runs use a response-grouped 26/6/6 train/validation/test split.

Schoenfeld boundary-probe design. The probe workflow excludes the 38 synthetic `</think>` closing units, which are all labelled Monitor, leaving the paper-matched 3,087 units. Qwen receives each SAT instruction as the user message and the released response as a teacher-forced assistant continuation. For a target unit, the feature is the hidden state immediately before its first token, so the probe sees the problem and preceding response but cannot read the target sentence. Hidden-state index 0 is the embedding output and indices 1–40 are the successive language-model layers.

At each index, seven separate target-versus-rest logistic regressions use raw 2,048-dimensional activations, balanced class weights, L2 regularization with $C = 1$, and no standardization or PCA. The paper-matched

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DOI:

protocol uses a target-stratified 80/20 sentence split with seed 42. This is a replication choice rather than a strong generalization design: all 38 response IDs occur in both partitions. The reported layer for each target maximizes accuracy on that same test split. For qualitative transfer inspection, the selected probes are applied to 20 examples each from GSM8K, MATH, PRM800K, and ProcessBench. The seven binary scores are independently trained and uncalibrated; their argmax is an inspection label, not a multiclass probability or new gold annotation. Only boundary vectors are stored; Qwen does not regenerate the released or benchmark solutions.

Correlation-pipeline design. The pipeline separates llama.cpp generation from a teacher-forced Hugging Face forward pass. It stores router logits, selected expert identifiers, and optional hidden states, then computes trace-level and repeated-attempt correlations with correctness and lexical reasoning-event labels. The current run covers 6,627 scored traces from 1,745 source problems across MATH-500, AIME 2024, AIME 2025, OlympiadBench, AMC 2023, Minerva Math, and GPQA-Diamond. AIME and AMC use 32 attempts per problem, GPQA uses ten attempts with permuted answer choices, and the other datasets use one. MMLU-Pro, the eighth and last default benchmark, was not run. Generation uses the UD-Q4_K_XL GGUF with embedded native MTP speculative weights; replay uses the matching BF16 checkpoint with runtime 4-bit quantization, so the stages use different 4-bit formats. All 6,627 traces are scored, and the repeated AIME and AMC designs support problem-cluster and within-problem analyses.

Table model and metric conventions. Every table caption identifies the evaluated model or models and the metrics shown. For classification, accuracy is the fraction of exact label matches, balanced accuracy is the unweighted mean of class recalls, precision and recall have their standard true-positive definitions, macro-F1 is the unweighted mean of per-class F1, and AUROC measures score ranking across the positive and negative classes. Cohen's κ is chance-corrected nominal agreement, Kendall's τ_b is rank agreement under the fixed episode-label ordering, and the GEPA composite averages their equal-weight $[0, 1]$ rescalings. Routing entropy is Shannon entropy of the softmax router distribution; switch rate is the fraction of top- k experts replaced from the preceding token; top- k overlap is adjacent-token Jaccard overlap; and router margin is the top-1 minus top-2 router probability. Unless a caption states otherwise, N and support are observation counts, r is a Pearson-family correlation coefficient, and confidence intervals are percentile bootstrap intervals.

Artifact provenance. The active packages are located at `src/moe_exp/gepaLLMAsJudge`, `src/moe_exp/probeTest`, and `src/moe_exp/correlation_pipeline`. GEPA runs through `run_gepaLLMAsJudge.sh` and writes to `results/gepaLLMAsJudge`; boundary-probe outputs remain under `results/probeTest`. The correlation workflow is launched by its local `run_all.sh` and uses its documented output directory.

The Qwen judge artifacts formerly called Experiment 0a, including the final Qwen3.8 model-alignment run, and the Qwen3.5 boundary-probe artifacts formerly called Experiment 0b are complete. Correlation generation, replay, and analysis are complete for seven of eight default benchmarks; only MMLU-Pro remains unrun. Historical OLMoE and numbered-experiment outputs are archived and are not pooled with the estimates in this report.

3 gepaLLMAsJudge

The gepaLLMAsJudge workflow evaluates whether prompt optimization improves a local judge for the seven adapted Schoenfeld episode labels: Read, Analyze, Plan, Implement, Explore, Verify, and Monitor. The model was the Q4_K_XL quantization of Qwen3.6-27B served by llama.cpp on one RTX 3090. All splits preserve complete responses, preventing neighboring units from crossing partitions.

Preliminary single-split runs. With seed 42, the 38 responses were divided into 26 training, 6 validation, and 6 test documents (2,382/407/336 units). These first runs classified the current unit without neighboring context and optimized per-unit exact match.

Table 1. Preliminary gepaLLMAsJudge agreement. The composite rescales Cohen’s κ and Kendall’s τ_b from $[-1, 1]$ to $[0, 1]$ and averages them. Coverage was 100% in every row. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: exact-match accuracy, Cohen’s κ , Kendall’s τ_b , their rescaled composite, and valid-output coverage.

Evaluation	Accuracy	κ	τ_b	Composite
Seed validation	0.6388	0.5581	0.4810	0.7598
Optimized validation	0.6855	0.6074	0.5847	0.7980
Optimized test	0.6607	0.5710	0.6066	0.7944
Few-shot validation	0.6806	0.6078	0.5830	0.7977
Few-shot test	0.6994	0.6178	0.6667	0.8211

The matched few-shot run used the same seed, document split, model, and GEPA budget, adding seven gold training examples to the instruction. It performs better than the optimized base condition on the held-out test set: accuracy rises from 66.07% to 69.94% (+3.87 percentage points), Cohen’s κ from 0.5710 to 0.6178, Kendall’s τ_b from 0.6066 to 0.6667, and composite agreement from 0.7944 to 0.8211. GEPA itself does not improve the few-shot validation result: the seed and selected optimized prompts both score 68.06% accuracy. The current comparison therefore favors few-shot prompting for held-out generalization, whereas the measurable GEPA optimization gain is confined to the base-prompt condition.

Context-aware nested cross-validation. The revised protocol excludes the same six responses from model selection and runs five outer folds over the remaining 32 responses. Within each fold, five responses form the inner validation set and all others outside the outer fold form the GEPA training set. Each request includes the problem statement, previous unit, current unit, and next unit. The seed instruction includes 21 audited synthetic contrastive demonstrations. GEPA receives inverse-frequency weighted exact match, whose split mean is balanced accuracy; the safety selector compares either balanced accuracy or macro-F1 and rejects any candidate whose class recall falls by more than the configured threshold.

Informative same-model reward. The current implementation can replace the static GEPA reward with an LLM-as-a-judge call to the same Qwen3.6-27B model. For each candidate label, the judge returns a short free-text explanation plus four diagnostic scores from 1 to 5: gold agreement, functional fit, contextual coherence, and boundary precision. Their normalized weighted mean is used only as GEPA’s optimization signal, with respective weights 0.40, 0.25, 0.15, and 0.20. Gold-label accuracy, balanced accuracy, macro-F1, and per-class recall remain the independent selection and reporting criteria. The judge runs deterministically with model reasoning disabled. Every judge call is persisted as JSONL with its component scores, normalized score, and free-text explanation, permitting a direct audit of feedback quality. Tables 2 and 3 remain static-reward cross-validation baselines; the completed LLM-judge final fit is reported separately below because its reused validation set is a prompt-development set rather than an outer-fold estimate.

Macro-F1 selection with the light budget is the strongest cross-validated configuration, but its advantage over balanced-accuracy selection is small: +0.32 accuracy points, +0.43 balanced-accuracy points, and +0.24 macro-F1 points. At the response level it wins on 15 documents, loses on 13, and ties on 4, so the observed gain is not stable across documents. The light run selects a GEPA-optimized prompt in three of five folds; the recall gate retains the seed prompt in the other two because Plan or Monitor recall falls too far.

Increasing the budget does not help. The heavy run selects an optimized prompt in only two folds and is 0.65 accuracy points and 0.68 macro-F1 points below the light run. Paired predictions show 38 units correct only under light versus 20 correct only under heavy. This difference is concentrated in fold 5, where light discovers and selects a strong candidate whereas heavy retains the seed after its candidate violates the Analyze recall gate.

Table 2. Aggregate selected-prompt outer-fold results over 2,789 units. “Balanced light” is the earlier 0.10 recall-gate run; the macro-F1 runs use a 0.05 gate. The six excluded responses were not evaluated. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, and the rescaled κ/τ_b composite.

Selection / budget	Accuracy	Balanced acc.	Macro-F1	Composite
Balanced / light	0.6701	0.6753	0.6480	0.8031
Macro-F1 / light	0.6734	0.6796	0.6505	0.8063
Macro-F1 / heavy	0.6669	0.6737	0.6437	0.8030

Table 3. Per-class outer-fold performance for the best current configuration, macro-F1 selection with the light GEPA budget. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: class support, precision, recall, and F1.

Class	Support	Precision	Recall	F1
Read	318	0.798	0.770	0.784
Analyze	685	0.757	0.632	0.689
Plan	205	0.524	0.590	0.555
Implement	650	0.880	0.722	0.793
Explore	193	0.679	0.492	0.571
Verify	538	0.669	0.606	0.636
Monitor	200	0.364	0.945	0.526

Independent heavy search is therefore not a superset of light search and can discard a better prompt found by a smaller run.

The revised objective improves overall balance only marginally and does not resolve Monitor overprediction. The model emits Monitor 519 times for 200 gold Monitor units; the leading errors are Verify→Monitor (127), Plan→Monitor (65), and Explore→Monitor (42). The annotation audit flags 180/3,125 units (5.76%), including 145 multiline units and 49 structural/control markers such as `</think>` and “Final Answer.” These artifacts plausibly inflate Monitor recall while depressing precision and should be reviewed before further optimization.

All runs retain 100% final-output coverage and complete without OOM failures, but the light and heavy jobs log 31 and 28 truncated LM responses during optimization. Moreover, the outer folds currently evaluate only the safety-selected prompt, not both seed and optimized prompts, so they do not directly estimate GEPA’s paired outer-fold gain. Finally, the six responses excluded from current CV were inspected in the preliminary experiments; they are locked from the current code path but no longer constitute a pristine research holdout. The judge should therefore not replace validated annotation until structural artifacts are adjudicated, decoding is constrained, paired outer evaluation is added, and a genuinely unseen external test set is designated.

Completed LLM-judge final fit. The August 11, 2026 run uses seed 42, the 21-example context-aware few-shot prompt, and the same-model informative reward on the original response-grouped 26/6/6 split (2,382/407/336 units). Seed and optimized prompts both return a valid label for all 407 validation units. Table 4 reports exact gold-label metrics rather than the LLM judge’s diagnostic score.

The optimized prompt fixes 49 seed errors and regresses on 14 previously correct units, a net gain of 35. All six validation responses improve, with response-level accuracy gains from 2.0 to 13.8 percentage points. Its primary effect is to curb the seed prompt’s overuse of Monitor: Monitor predictions fall from 74 to 43, including nine corrections from Monitor to true Plan and 15 from Monitor to true Verify. Recall rises most for Plan (+28.13

Table 4. Context-aware LLM-judge final-fit results on the 407-unit prompt-development validation set. Coverage is 100% in both rows. The locked test is not evaluated. *Model(s)*: Qwen3.6-27B Q4_K_XL, used both for episode-label prediction and the same-model GEPA reward. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, Cohen’s κ , Kendall’s τ_b , absolute change, and valid-output coverage.

Prompt	Accuracy	Balanced acc.	Macro-F1	κ	τ_b
21-example seed	0.6683	0.6765	0.6132	0.5988	0.5904
GEPA optimized	0.7543	0.7769	0.7270	0.6968	0.6832
Absolute change	+0.0860	+0.1003	+0.1138	+0.0980	+0.0929

Table 5. Per-class recall and optimized F1 for the completed LLM-judge final fit. *Model(s)*: Qwen3.6-27B Q4_K_XL, used both for episode-label prediction and the same-model GEPA reward. *Metrics*: class support, seed and optimized recall, and optimized per-class F1.

Class	Support	Seed recall	Opt. recall	Opt. F1
Read	45	0.778	0.756	0.791
Analyze	122	0.598	0.680	0.744
Plan	32	0.531	0.813	0.722
Implement	103	0.786	0.825	0.825
Explore	17	0.471	0.706	0.686
Verify	70	0.571	0.714	0.763
Monitor	18	1.000	0.944	0.557

points), Explore (+23.53), Verify (+14.29), and Analyze (+8.20). Monitor recall falls slightly from 1.000 to 0.944, but precision rises from 0.243 to 0.395. The selected prompt still has weak Monitor precision and Analyze recall, and the remaining errors concentrate at the Analyze/Implement/Verify boundaries and Verify→Monitor. Implement is the only class whose F1 decreases, from 0.839 to 0.825, because its additional predictions include more false positives.

The internal same-model reward rises from 0.7264 for the seed to 0.8196 for the selected candidate. The optimization trace contains 145 candidates, 144 full validation evaluations, and 60,767 persisted judge calls, of which 148 are judge-output errors. A text audit finds 37 validation units for which the judge explicitly disputes the corpus gold label in at least one evaluation, showing that the reward itself detects annotation-sensitive boundaries rather than consistently treating gold as semantically authoritative. The structural audit flags 29 validation units; their accuracy remains 72.4% under both prompts, so the gain comes entirely from otherwise unflagged units.

GEPA expands the instruction from 21 to 40 demonstrations and from 7,648 to 32,717 characters. The extra rules and examples target observed corpus edge cases, raising inference cost and the risk of learning annotation conventions rather than a portable episode taxonomy. More importantly, GEPA evaluates 145 candidates against the same six-response validation set used for final selection. The nominal 407 units are also correlated within responses, and only one optimization seed is available. The result files explicitly record `locked_test_evaluated: false` and `test: null`. The observed gain is therefore strong evidence of successful prompt optimization on the development distribution, but not a held-out generalization estimate. The prompt should be frozen and compressed or ablated before a one-time test on genuinely unseen responses.

Table 6. Final Qwen3.8 model-alignment results on the 407-unit development validation set. Both prompts have 100% valid-output coverage; the 336-unit locked test is not evaluated. *Model(s)*: Qwen3.8-27B UD-Q4_K_XL, used for both classification and the same-model GEPA reward. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, Cohen’s κ , and Kendall’s τ_b .

Prompt	Accuracy	Balanced acc.	Macro-F1	κ	τ_b
Imported 40-example seed	0.7174	0.7400	0.6938	0.6523	0.6122
GEPA optimized	0.7125	0.7288	0.6743	0.6475	0.6009
Absolute change	−0.0049	−0.0112	−0.0196	−0.0049	−0.0113

Table 7. Per-class recall in the final Qwen3.8 alignment. The selection gate rejects any candidate whose recall drops by more than 0.10 for any class. *Model(s)*: Qwen3.8-27B UD-Q4_K_XL. *Metrics*: class support and seed versus GEPA-optimized recall.

Class	Support	Seed recall	Opt. recall
Read	45	0.778	0.756
Analyze	122	0.549	0.533
Plan	32	0.750	0.563
Implement	103	0.874	0.874
Explore	17	0.529	0.647
Verify	70	0.700	0.786
Monitor	18	1.000	0.944

Final Qwen3.8 model-alignment pass. The August 27–28, 2026 alignment run uses Qwen3.8-27B UD-Q4_K_XL, seed 42, the heavy GEPA budget, and medium reasoning effort for the same-model judge. It keeps the same response-grouped 2,382/407/336 train/validation/locked-test split. Rather than drawing new few-shot examples, it imports the previous 40-example Qwen3.6 optimized prompt as its external seed. Gold-label agreement supplies 40% of the internal judge reward; functional fit, contextual coherence, and boundary precision supply the remaining 60%. Final selection uses exact balanced accuracy with a maximum allowed per-class recall drop of 0.10.

The optimized candidate changes 49 labels: it fixes 17 seed errors, breaks 19 seed-correct cases, and changes 13 errors to other errors. Its largest failure is Plan recall, which falls by 0.188 and therefore breaches the recall gate; the selected prompt is byte-identical to the imported seed. An unclustered exact McNemar test on the 17 versus 19 discordant correctness outcomes gives $p = 0.868$, offering no evidence of an aggregate improvement, although this unit-level calculation understates response clustering.

This negative exact-label result contrasts with the internal GEPA objective, whose best candidate reward rises from 0.7684 to 0.7762. Because only 40% of that reward is gold agreement, the discrepancy is consistent with optimizing the judge’s broader functional criteria without improving corpus-label classification. The optimized instruction is shorter (2,591 versus 5,201 words) but internally inconsistent: an early rule treats “Let me compute that” as Monitor while a later rule treats “Let me calculate” as Plan. Plan-to-Monitor errors rise from 6 to 10, consistent with that boundary confusion. The best internal candidate appears by iteration 3 and remains the best through iteration 19, so the remaining heavy search adds roughly 14.5 hours without a better candidate. The appropriate conclusion is that the final alignment completed correctly and the safety gate worked: GEPA found no safe improvement over the existing seed. The complete artifact is stored at `results/gepaLLMAsJudge/qwen3.8-27b-medium-final-s42-v3`.

Table 8. Accuracy-selected Schoenfeld boundary probes on the sentence-level test split. Prevalence is over all 3,087 units. Δ Maj. subtracts the always-negative accuracy. Balanced accuracy is derived from the saved test confusion matrix. Layer selection and evaluation use the same test split and are therefore selection-optimistic. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen hidden-state encoder, with one-versus-rest logistic probes. *Metrics*: target prevalence, selected hidden-state layer, accuracy, accuracy minus the majority baseline, balanced accuracy, F1, and AUROC.

Target	Prevalence	Layer	Accuracy	Δ Maj.	Balanced acc.	F1	AUROC
Read	12.21%	34	0.9239	+0.0461	0.8476	0.7044	0.9338
Analyze	25.62%	36	0.8107	+0.0669	0.7669	0.6465	0.8348
Plan	7.29%	39	0.9175	-0.0096	0.7405	0.4848	0.8551
Implement	21.87%	38	0.8964	+0.1151	0.8617	0.7714	0.9305
Explore	6.74%	40	0.9223	-0.0103	0.7486	0.4894	0.8884
Verify	20.38%	27	0.8754	+0.0792	0.8509	0.7260	0.9122
Monitor	5.90%	35	0.9304	-0.0106	0.7546	0.4819	0.9031

4 probeTest

The probeTest workflow tests a different annotation mechanism from the generative judge in gepaLLMAsJudge. It asks whether Qwen’s representation of the prefix immediately before a sentence contains linearly accessible information about the upcoming Schoenfeld episode. It therefore measures prospective prefix-state decodability, not semantic classification after reading the sentence and not the latent dynamics of a Qwen-generated trace.

Within-corpus results. Table 8 reports the accuracy-selected layer for each one-versus-rest target. AUROC is 0.835–0.934 and balanced accuracy is 0.740–0.862, demonstrating substantial within-split ranking signal. Implement is strongest at the natural threshold (F1 0.771), followed by Verify (0.726) and Read (0.704). Analyze is intermediate. Plan, Explore, and Monitor have F1 below 0.50. Their nominal accuracies are also 0.96–1.06 percentage points below an always-negative majority classifier, so accuracy alone gives a misleading impression for these rare targets.

The signal emerges rapidly and is not maximal at the model output. Mean one-versus-rest AUROC across targets rises from 0.624 at the embedding output to 0.873 at layer 10 and 0.892 at layer 20, peaks at 0.899 at layer 30, and falls to 0.867 at layer 40. Mean F1 peaks at layer 28. Accuracy-based target selection sometimes obscures this profile: for Plan, the selected layer 39 has AUROC 0.855 whereas layer 21 reaches 0.922. AUROC or balanced accuracy, chosen inside a validation fold, better matches the scientific question of decodability.

The fits are numerically stable—none of the 287 regressions reports a convergence warning—but statistical overfitting remains visible. At the selected layers, train–test accuracy gaps are 16.7 points for Analyze, 10.1 for Implement, approximately 7.8 for Plan and Explore, and 4.5–5.5 for the other targets. Explore reaches 100% training accuracy. With 2,048 features, 2,469 training units, strong within-response dependence, and only 38 distinct responses, the effective independent sample size is much closer to the number of responses than to the number of sentences. Test-based layer selection adds further optimism.

External benchmark inspection. All 80 requested examples complete without omitted units, yielding 581 sentence boundaries. Table 9 shows that the seven-way inspection behavior collapses mainly to Implement, Read, and Monitor. Implement alone receives 49.2% of argmax assignments, whereas Explore is never selected. The transfer distribution also differs sharply from training: Analyze falls from 25.6% of gold units to 7.7% of predictions and Verify from 20.4% to 3.3%.

Table 9. Argmax inspection-label counts on external reasoning units. These are raw prediction frequencies, not accuracies. Table 10 compares them with the subsequent provisional pragmatic review. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: unit counts and seven-way argmax prediction counts.

Dataset	Units	Read	Analyze	Plan	Implement	Explore	Verify	Monitor
GSM8K	96	0	2	3	53	0	0	38
MATH	88	23	7	1	45	0	1	11
PRM800K	139	34	18	0	71	0	1	15
ProcessBench	258	69	18	12	117	0	17	25
Total	581	126	45	16	286	0	19	89

Table 10. Agreement with the provisional pragmatic review. Primary accuracy requires the preferred label; compatible accuracy accepts either label on a multi-label unit. Macro-F1 and macro recall treat each accepted label as a positive reference membership. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: unit count, primary accuracy, compatible accuracy, multi-label count, accepted-membership macro-F1, and macro recall.

Dataset	Units	Primary acc.	Compatible acc.	Multi-label	Macro-F1	Macro recall
GSM8K	96	49.0%	62.5%	20	0.136	0.129
MATH	88	40.9%	53.4%	36	0.211	0.228
PRM800K	139	46.0%	49.6%	44	0.241	0.302
ProcessBench	258	56.2%	68.6%	87	0.396	0.350
Total	581	50.3%	60.8%	187	0.330	0.306

The independent binary scores do not form a clean multiclass decision rule. Seventy-four units (12.7%) have no score above 0.5 but are still assigned an argmax label, while 180 (31.0%) activate more than one probe and 99 (17.0%) have a top-two margin below 0.1. A conservative display rule requiring a top score of at least 0.8 and a margin of at least 0.2 retains only 318/581 units (54.7%), mostly Implement, Read, and Monitor. This threshold is a useful abstention heuristic, not a calibrated guarantee of correctness.

Provisional pragmatic adjudication. To make the qualitative inspection measurable, every displayed unit was subsequently reviewed using the problem, the current unit, its neighbors, and the adapted Schoenfeld precedence rule. The annotation is a hybrid rule-assisted Codex pragmatic review, not human consensus or authoritative benchmark gold. It records one primary label for conventional single-label statistics and, where two functions are genuinely defensible, an accepted set containing both labels. This occurs for 187/581 units (32.2%), mostly at boundaries where a unit both announces and executes an operation or where a structural marker also carries a direct result. Of the annotations, 394 have high, 150 medium, and 37 low confidence; the low-confidence cases remain priority items for human review.

The probe agrees with the primary pragmatic choice on 292/581 units (50.3%) and with at least one accepted label on 353/581 (60.8%). Thus recognizing genuine taxonomy overlap recovers 61 additional compatible decisions, but it does not make the reference authoritative. ProcessBench has the strongest compatible agreement (68.6%), followed by GSM8K (62.5%); MATH has the lowest strict agreement (40.9%). Table 10 reports the complete dataset-level results.

Table 11. Per-label comparison with accepted pragmatic memberships. Support counts every unit for which the label is accepted, including secondary labels on ambiguous units. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: accepted-label support, argmax prediction count, precision, recall, and F1.

Label	Accepted support	Probe predictions	Precision	Recall	F1
Read	84	126	0.437	0.655	0.524
Analyze	84	45	0.356	0.190	0.248
Plan	90	16	0.375	0.067	0.113
Implement	345	286	0.804	0.667	0.729
Explore	11	0	0.000	0.000	0.000
Verify	51	19	0.632	0.235	0.343
Monitor	103	89	0.382	0.330	0.354

The class-level comparison in Table 11 sharpens the interpretation. Implement is the only operationally strong external label (precision 0.804, recall 0.667, F1 0.729). Read is moderately useful (F1 0.524). Analyze and Plan are severely under-produced, while Verify and Monitor have low recall. Explore has 11 accepted reference memberships but is never predicted, hence zero recall. Because multi-label units contribute to two reference classes, accepted supports overlap and do not sum to 581.

Pragmatically, Implement is the most coherent transferred label: it marks long calculation, substitution, and algebra runs. Read often marks givens or restatements, particularly near the start of MATH and PRM800K solutions. Monitor behaves primarily as a solution-boundary or opening-state detector: all 20 GSM8K first units and 18/20 ProcessBench first units receive Monitor, whereas 19/20 MATH and 17/20 PRM800K first units receive Read. Because the target sentence is unseen at this point, this source dependence must arise from the problem, prefix position, and formatting rather than the displayed unit’s semantics. Analyze and Plan appear plausibly on some explanatory or phase-change boundaries but have low coverage. Explore is unusable for external tagging in the present model.

Verify has a narrower but potentially useful role as a review or reconsideration flag. All 17 ProcessBench Verify predictions occur at or after the benchmark’s annotated first error, including six within one trace that repeatedly reinterprets the question before ending with the wrong sign. This is not an error detector: errors generally occur early, Verify may merely mark late position or reflective language, and no PRM800K post-error unit is labelled Verify. It can prioritize passages for inspection but cannot judge correctness.

The qualitative stage also exposes an input problem independent of the classifier. Thirteen units are split immediately after abbreviations such as “Mr.”, producing fragments including “Therefore, Mr.” and inconsistent scores. External use should first adopt abbreviation- and LaTeX-aware segmentation. With the current outputs, the defensible practical representation is a coarse three-phase view—setup (Read and opening Monitor), execution (Analyze, Plan, and Implement), and review (Verify)—plus an explicit unknown/ambiguous state. The seven raw argmax labels should not be used as automatic cognitive annotations.

5 Qwen Correlation Pipeline

5.1 Seven-benchmark run

The September 2, 2026 run completed generation, teacher-forced routing replay, and cluster-aware analysis for seven of the eight default SPIRAL benchmarks. It contains 6,627 scored traces from 1,745 source problems. MMLU-Pro, the last benchmark in the default order, was not run and is omitted from every total and association

Table 12. Coverage, answer accuracy, and lexical-event prevalence in the completed seven-benchmark run. Accuracy is the per-trace mean: pass@1 for single-attempt datasets, avg@32 for AIME and AMC, and avg@10 for GPQA-Diamond. The pooled row is descriptive only and is not an overall benchmark score. *Model(s)*: Qwen3.5-35B-A3B (UD-Q4_K_XL GGUF generation and matching unsloth/Qwen3.5-35B-A3B routing replay with runtime 4-bit quantization). *Metrics*: problem and trace counts, exact/symbolic or multiple-choice answer accuracy, and lexical backtracking, contradiction, and self-correction rates.

Dataset	Problems	Traces	Correct	Accuracy	Backtrack	Contradict	Self-correct
MATH-500	500	500	403	0.806	127 (25.4%)	63 (12.6%)	81 (16.2%)
AIME 2024	30	960	158	0.165	653 (68.0%)	295 (30.7%)	508 (52.9%)
AIME 2025	30	960	141	0.147	579 (60.3%)	255 (26.6%)	449 (46.8%)
OlympiadBench	675	675	280	0.415	319 (47.3%)	212 (31.4%)	226 (33.5%)
AMC 2023	40	1,280	613	0.479	580 (45.3%)	360 (28.1%)	428 (33.4%)
Minerva Math	272	272	107	0.393	117 (43.0%)	34 (12.5%)	86 (31.6%)
GPQA-Diamond	198	1,980	301	0.152	1,466 (74.0%)	432 (21.8%)	1,115 (56.3%)
Pooled	1,745	6,627	2,003	0.302	3,841 (58.0%)	1,651 (24.9%)	2,893 (43.7%)

Table 13. Prespecified trace-level correlations with correctness. Entries are point-biserial r values; bold entries have 95% problem-cluster bootstrap intervals that exclude zero. These are marginal, not length-adjusted, associations. GPQA attempts also permute answer positions. *Model(s)*: Qwen3.5-35B-A3B, generated and replayed as in Table 12. *Metrics*: point-biserial correlation between correctness and prespecified aggregate trace features.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva	GPQA
Token count	-0.477	-0.637	-0.673	-0.618	-0.640	-0.356	-0.690
Router confidence	0.513	0.310	0.552	0.609	0.587	0.036	0.224
Router top-1/top-2 margin	0.234	0.089	0.341	0.369	0.293	-0.178	0.038
Hidden-state norm	-0.477	-0.357	-0.270	-0.503	-0.462	-0.019	0.065
Hidden-state step distance	-0.295	-0.187	-0.291	-0.428	-0.541	-0.217	0.058
Hidden/router geometry	-0.190	0.098	-0.163	-0.196	-0.256	-0.218	-0.245

below. The run uses one attempt per problem on MATH-500, OlympiadBench, and Minerva Math; 32 attempts per problem on AIME 2024, AIME 2025, and AMC 2023; and ten attempts with deterministically permuted answer choices on GPQA-Diamond.

Generation uses Qwen3.5-35B-A3B at temperature 0.6 and top- p 0.95 through the UD-Q4_K_XL GGUF with its embedded native-MTP weights. The saved messages are replayed through the matching Hugging Face checkpoint with runtime 4-bit quantization. Compact features are computed at the six router layers selected independently by probeTest: 27, 34, 35, 36, 38, and 39. The analysis exposes 83 features and uses 500 problem-cluster bootstrap replicates for the prespecified aggregate features.

The benchmark accuracies should not be compared as if they shared a common population: they differ in subject, difficulty, response format, and number of attempts. In particular, the 0.302 pooled fraction gives repeated-attempt benchmarks more weight and is not the unweighted mean of benchmark scores.

5.2 Trace-level correctness associations

Table 13 reports the prespecified marginal point-biserial associations. The most consistent result is response length: correct traces are shorter on all seven datasets, with r from -0.356 to -0.690 and all problem-cluster intervals excluding zero. Mean normalized router confidence is positively associated with correctness on six datasets ($r = 0.224$ – 0.609); Minerva is the exception ($r = 0.036$, 95% CI $[-0.081, 0.154]$). Reduced hidden-state step distance also has a negative association on six datasets, while GPQA-Diamond is null. Hidden-state norm is negative on the five mathematics datasets other than Minerva, but null on Minerva and GPQA-Diamond.

Table 14. Within-problem correct-minus-incorrect contrasts for prespecified features. Intervals resample mixed-outcome source problems. Confidence and margin differences are small because the entries are token- and layer-averaged router probabilities. GPQA-Diamond is deliberately excluded because its ten attempts permute answer positions. *Model(s)*: Qwen3.5-35B-A3B, generated and replayed as in Table 12. *Metrics*: mean within-problem feature difference and 95% problem-bootstrap interval.

Dataset	Feature	Mixed problems	Difference	95% CI
AIME 2024	Token count	19	-362	$[-608, -125]$
	Router confidence	19	$+3.09 \times 10^{-4}$	$[1.72, 4.37] \times 10^{-4}$
	Router margin	19	$+4.52 \times 10^{-5}$	$[0.38, 8.58] \times 10^{-5}$
AIME 2025	Token count	15	-456	$[-818, -147]$
	Router confidence	15	$+4.26 \times 10^{-4}$	$[2.89, 5.58] \times 10^{-4}$
	Router margin	15	$+8.45 \times 10^{-5}$	$[3.50, 14.14] \times 10^{-5}$
AMC 2023	Token count	27	-636	$[-967, -323]$
	Router confidence	27	$+3.94 \times 10^{-4}$	$[2.08, 5.99] \times 10^{-4}$
	Router margin	27	$+7.53 \times 10^{-5}$	$[2.55, 14.40] \times 10^{-5}$

The broader run therefore revises the earlier preliminary interpretation. Router confidence is no longer a MATH-500-only result: its positive direction appears in six benchmarks and is nonzero under the cluster bootstrap in each of those six. It is still not universal. Minerva confidence is null, and its top-1/top-2 margin reverses direction. Transition metrics are also heterogeneous: correct MATH-500, AMC, and Olympiad traces generally switch experts less, whereas Minerva and GPQA show more switching and less adjacent top- k overlap. A single dataset-pooled coefficient would conceal these differences and is not treated as replication evidence.

5.3 Repeated-attempt analyses

The repeated designs now make comparisons within the same question possible. There are both correct and incorrect attempts for 19 of 30 AIME 2024 problems, 15 of 30 AIME 2025 problems, and 27 of 40 AMC 2023 problems. In all three datasets, the mean correct attempt is shorter and has slightly higher router confidence and top-1/top-2 margin than the mean incorrect attempt; all nine problem-bootstrap intervals in Table 14 exclude zero. Thus, differences in which problems were sampled do not by themselves explain these three effects.

Problem-level results agree with this pattern but remain exploratory. Mean token count has Spearman $\rho = -0.606$, -0.747 , and -0.845 with per-problem pass rate on AIME 2024, AIME 2025, and AMC 2023, respectively. Mean router confidence has $\rho = 0.228$, 0.569 , and 0.742 . On GPQA-Diamond, for which within-problem contrasts are excluded, the corresponding problem-level values are -0.789 and 0.360 . The analyzer retains naive problem-level p -values only as diagnostics and does not supply multiplicity-adjusted uncertainty for these rankings.

The within-problem result controls source-question identity, but it does not show that routing concentration contributes information beyond trace length or token composition. Correct attempts are hundreds of tokens shorter in the same comparisons. A length-conditioned, dataset-aware model is therefore needed before calling confidence an independent predictor, and observational correlation cannot identify a causal routing mechanism.

5.4 Lexical events and storage audit

Backtracking and self-correction markers are primarily markers of longer traces. In the pooled data their correlations with token count are 0.425 and 0.332 , and with router confidence are -0.256 and -0.205 . The length direction recurs dataset by dataset, but the confidence association is null on both AIME datasets and on Minerva. Contradiction is weaker and less stable. The event rates in Table 12 also vary sharply across datasets. These lexical detectors are descriptive trace-style indicators, not validated reasoning-episode annotations, and pooled event correlations partly encode benchmark identity.

The compact extraction design succeeds operationally at the larger scale. Across 49.93 million retained tokens, full selected-layer payloads would have required 1.534 TB (1.395 TiB). The run instead stores 4.81 GB of compact tensor artifacts and 5.30 GB of per-dataset files, while recording `raw_tensors_saved: false` and `expert_weights_saved: false`. Compact tensor storage is about 0.31% of the projected raw payload, avoiding more than 99.68% while preserving the analyzed scalar features and top- k expert identifiers.

The immediate remaining execution task is MMLU-Pro. The principal analysis task is a prespecified model that conditions on length and benchmark while respecting problem clusters; layerwise maxima should remain exploratory unless their selection and multiplicity are handled explicitly.

6 Analysis

6.1 Conclusions supported so far

- (1) The static-reward, context-aware GEPA study reaches 0.650 cross-validated macro-F1 with the light budget. The advantage over balanced-accuracy selection is small and the heavy search is worse. In the separate completed same-model LLM-judge final fit, development-set balanced accuracy rises from 0.677 to 0.777 and macro-F1 from 0.613 to 0.727, primarily through reduced Monitor overprediction. This establishes successful prompt optimization on the development distribution, not held-out generalization. The final Qwen3.8-27B alignment does not reproduce that gain: balanced accuracy falls from 0.740 for the imported seed to 0.729 for the GEPA candidate, and the prespecified recall gate correctly retains the seed.
- (2) The boundary probes show substantial sentence-split decodability of upcoming Schoenfeld episodes, with selected-layer AUROC of 0.835–0.934 and the strongest thresholded performance for Implement, Verify, and Read. This establishes accessible prefix-state signal within familiar responses, not generalization to unseen reasoning traces. External application is best interpreted as coarse phase inspection: a provisional pragmatic review gives 50.3% strict and 60.8% compatible agreement, with 187/581 multi-label units. Implement is strongest (F1 0.729), while the remaining labels are incomplete external tags. ProcessBench Verify predictions identify review-heavy post-error passages but do not identify correctness.
- (3) The correlation run completes 6,627 scored traces from 1,745 problems on seven of eight default benchmarks; only MMLU-Pro is unrun. Correct traces are shorter on every dataset ($r = -0.356$ to -0.690). Normalized router confidence is positively associated with correctness on six datasets ($r = 0.224$ – 0.609) and null on Minerva. More importantly, correct attempts have higher confidence within the same mixed-outcome AIME 2024, AIME 2025, and AMC 2023 problems, with all three problem-bootstrap intervals excluding zero. Those correct attempts are also 362–636 tokens shorter on average, so confidence has not yet been shown to add information beyond length.

6.2 What has not been established

Neither GEPA final fit provides an unbiased test estimate. The earlier run selects among 145 candidates on the same six-response validation set, has only one optimization seed, and leaves the locked 336-unit test unevaluated. The result does not establish that the expanded 40-demonstration prompt transfers to new responses or that the same-model reward is an independent evaluator. The Qwen3.8 alignment reuses that selected prompt as its seed, optimizes and selects on the same validation partition, and also leaves the locked test untouched; retaining the seed is evidence that this search found no safe development-set improvement, not evidence of generalization.

The boundary-probe run does not establish response-grouped generalization, calibrated seven-way episode classification, or the cognitive states of Qwen’s own generations. It uses Qwen to encode teacher-forced DeepSeek-R1 text, shares all 38 response IDs between train and test, and selects layers on the reported test set. Its external argmax outputs have no authoritative human or benchmark episode ground truth. Agreement with the provisional hybrid review is diagnostic and should not be called validated transfer accuracy.

The correlation results do not establish a dataset-invariant routing predictor or a causal mechanism. No length-conditioned or dataset-adjusted model has been fit. The repeated-attempt comparisons control question identity but not length or token content, GPQA within-problem contrasts are excluded because answer positions are permuted, and MMLU-Pro is unrun. Raw pooled coefficients mix datasets with different accuracy, trace length, and routing baselines, and layerwise maxima are exploratory comparisons without selection-adjusted uncertainty. Across all three workflows, correlation and linear decodability should not be read as causal evidence.

7 Discussion

The completed GEPA final fit demonstrates that same-model informative feedback can guide a large prompt toward the development labels: it fixes 49 seed errors while introducing 14 regressions and improves every validation response. Its optimization trace also shows the central risk. The prompt expands from 21 to 40 demonstrations and is selected after 145 candidate evaluations on those same six responses. The next defensible test is therefore to freeze and ablate or compress the prompt, quantify response-level uncertainty, and evaluate it once on genuinely unseen responses.

The Qwen3.8 alignment is an important negative result. Although its internal same-model reward increases, exact corpus-label metrics decline and Plan recall breaches the prespecified 0.10 safety threshold. Its shorter optimized instruction also contains conflicting guidance for phrases such as “Let me compute that,” plausibly contributing to more Plan-to-Monitor errors. The safety gate therefore performs its intended role: optimization reward alone is insufficient evidence for deployment, and the byte-identical selected seed should now be frozen rather than optimized further on this validation set.

The episode probes add a complementary representation-level result: the prefix before a sentence often encodes the phase likely to follow, and the signal is strongest in middle-to-late layers rather than at the final model output. The current split cannot determine whether this reflects a portable episode representation, trace-specific transition regularities, position, or formatting. A response-grouped nested evaluation should select both layer and regularization on inner folds, evaluate once on unseen responses, repeat over seeds, and report response-bootstrap intervals. Standardized or norm-matched features and simple position, previous-unit, and formatting baselines are needed to separate hidden-state information from layer scale and trace structure.

For external use, the probe outputs should be treated as abstaining phase suggestions rather than annotations. Abbreviation- and LaTeX-aware segmentation should precede evaluation; independent one-versus-rest scores should either be calibrated on a common multiclass validation set or retain multiple labels. The pragmatic review confirms substantial ambiguity and only 60.8% compatible agreement. Independent human annotators should resolve the 37 low-confidence units and assess inter-annotator agreement before any transfer-accuracy claim. Cross-model recognition, as studied here, should also be separated from Qwen’s own generation by running the protocol on Qwen-generated traces.

The seven-benchmark correlation run strengthens the router-confidence result without making it universal. The positive marginal association appears in six datasets and survives within-question comparison on AIME 2024, AIME 2025, and AMC 2023. Minerva remains null, and its margin and expert-transition directions differ from several other benchmarks. The even more consistent finding is that correct responses are shorter, including within the repeated problems. Pooled coefficients and lexical-event correlations remain liable to dataset and trace-style confounding. The next defensible analysis should condition confidence on length and token composition, include dataset effects, and retain problem-cluster uncertainty.

8 Conclusion

The three active workflows support deliberately limited claims. GEPA improves the earlier same-model-judge development result substantially, but the final Qwen3.8 alignment finds no safe improvement over that imported

prompt and neither run has been tested on an untouched response set. The boundary probes reveal strong within-split linear access to upcoming episode labels, especially for Implement, Verify, and Read, but their sentence split leaks every response across train and test and their external outputs collapse toward a coarse setup, execution, and review taxonomy. The seven-benchmark correlation study finds a robust association with shorter correct traces and a positive router-confidence association on six datasets and within three repeated-problem designs. Minerva, heterogeneous transition metrics, and co-occurring length effects prevent a dataset-invariant, independent, or causal routing claim.

The immediate evidential requirements are a frozen-prompt GEPA evaluation on unseen responses, response-grouped nested boundary probes with independently validated external labels, completion of MMLU-Pro, and a length-conditioned dataset-aware analysis of the repeated-attempt routing result.

9 Immediate Work

The agreed order of work is:

- (1) freeze and archive the seed prompt retained by the completed final GEPA alignment, and reserve the locked test for a prespecified evaluation;
- (2) run the remaining MMLU-Pro generation, replay, and analysis stages;
- (3) fit a prespecified length-conditioned, dataset-aware correctness model that respects problem clusters and keeps layerwise screening exploratory;
- (4) test whether the repeated-attempt router-confidence contrast survives controls for length and token composition; and
- (5) expand to additional MoE families only after the pilot pipeline and disk budget are stable.

The detailed acceptance criteria and transcript ambiguities are maintained in `NEXT_STEPS.md` at the repository root.

A Reproducibility Checklist

Table 15. Living inventory for the three active workflows. *Model(s)*: Qwen3.6-27B and Qwen3.8-27B Q4_K_XL, Qwen/Qwen3.5-35B-A3B-GPTQ-Int4, and Qwen3.5-35B-A3B correlation-run variants. *Metrics*: workflow status, sample counts, headline metrics, and next actions; this is an inventory rather than a common-metric evaluation.

Item	Status	Note / next action
gepaLLMAsJudge	Cross-validation and two final-fit validations complete; test pending	Static-reward grouped cross-validation over 2,789 units gives 67.34% accuracy and 65.05% macro-F1 for light-budget GEPA; heavy is worse. The same-model LLM-judge final fit improves validation accuracy from 66.83% to 75.43% and macro-F1 from 0.613 to 0.727, but reuses six responses across 145 candidate evaluations. In the final Qwen3.8 alignment, GEPA instead reduces balanced accuracy from 0.740 to 0.729 and Plan recall from 0.750 to 0.563, so the safety gate correctly retains the imported 40-example seed. Neither run evaluates the 336-unit locked test. Freeze the selected seed and evaluate it only under a prespecified held-out protocol.
probeTest	Complete; grouped and human validation pending	Qwen3.5 GPTQ probes cover 3,087 units, seven targets, and 41 hidden-state indices; selected-layer AUROC is 0.835–0.934. A provisional hybrid review of 581 external units gives 50.3% primary and 60.8% compatible agreement, with 187 multi-label and 37 low-confidence units; Explore is never predicted. Grouped cross-validation, robust segmentation, multiclass calibration, and independent human adjudication remain pending.
correlation_pipeline	Seven benchmarks complete; MMLU-Pro pending	Generation, replay, and analysis cover 6,627 scored traces from 1,745 problems on MATH-500, AIME 2024/2025, OlympiadBench, AMC 2023, Minerva Math, and GPQA-Diamond. Correct traces are shorter on all seven datasets. Router confidence correlates positively with correctness on six and is null on Minerva; it is also higher within mixed-outcome AIME and AMC problems, but those correct traces are hundreds of tokens shorter. Generation uses the UD-Q4_K_XL GGUF and replay uses the matching checkpoint with runtime 4-bit quantization. Run MMLU-Pro and test the confidence effect after conditioning on length and benchmark.